Thomas C. Fraser

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Curriculum Vitae

Personal

Email	tcf@math.ku.dk	
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Phone	DEN: +45 81 94 61 22	CAN: +1 226 868 0557
Address	Copenhagen, Denmark	
Citizenship	Canadian	
Date of Birth	27/11/1994	

Positions & education

Postdoctoral Researcher — Jan. 2024 – present
Department of Mathematics, University of Copenhagen

Ph.D., Physics — Sep. 2018 – Sep. 2023
Perimeter Institute For Theoretical Physics, Waterloo, ON
Thesis: An estimation theoretic approach to quantum realizability problems

M.Sc., Theoretical Physics, Quantum Information — Aug. 2017 – Jun. 2018
Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON
Thesis: A Combinatorial Solution to Causal Compatibility

B.Sc., Mathematical Physics, Astrophysics Specialization — Sep. 2012 – Apr. 2017
University of Waterloo, Waterloo, ON

Work experience

Research & Development, Machine Learning, Data Scientist — Sep. 2015 – Jan. 2016
Meltwater (formerly Sysomos), Toronto, ON, Canada

Video Game Developer — Sep. 2014 – May 2015
Lunarch Studios, Waterloo, ON, Canada

Mathematics Tutor — Jan. 2014 – May 2014
Humber College, Toronto, ON, Canada

Solar Panel Technician — Jun. 2011 – Aug. 2011
OVG Solar, Inc, Renfew, ON, Canada

Skills

Mathematics	Bayesian networks, causal modelling, directed acyclic graphs, belief propagation, large deviations theory, convex and non-convex optimization, group representation theory, geometric invariant theory, operator algebras, spectral theory of tensors
Programming	C, C++, Python, Mathematica, Lua, Matlab, Actionscript, JavaScript

Papers

2026 On the two-copy distillability of Werner states and a new partial trace inequality
Thomas C Fraser, Felix Huber, Balázs Pozsgay, István Vona
arXiv:2607.24309

2026 Complete entanglement detection using polynomial invariants
Thomas C. Fraser, V. Blakaj, R. Rubboli, F. Huber, M. Fanizza
arXiv:2606.16712

2026 On quantum functionals for higher-order tensors
A. Botero, M. Christandl, Thomas C Fraser, I. Leigh, H. Nieuwboer
arXiv:2604.18283

2025 The tetrahedral Horn problem and asymptotics of $U(n)$ $6j$ symbols
A. Alekseev, M. Christandl, Thomas C. Fraser
arXiv:2510.04877

2023 An estimation theoretic approach to quantum realizability problems
Thomas. C. Fraser
Ph.D. Thesis. arXiv:2402.10902

2023 Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory
D. Schmid, Thomas. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens
Quantum 7, 1194

2022 A sufficient family of necessary inequalities for the compatibility of quantum marginals
Thomas. C. Fraser
arXiv preprint arXiv:2211.00685

2020 A Combinatorial Solution to Causal Compatibility
Thomas. C. Fraser
Journal of Causal Inference 8(1), 22-53

2018 Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry
J. R. Rinehart, J. H. Béjanin, Thomas. C. Fraser, M. Mariantoni
arXiv preprint arXiv:1804.04756

2018 Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario
Thomas. C. Fraser, E. Wolfe
Physical Review A, 98(2), 022113

Teaching

Block 3 2025-2026 Introduction to Representation Theory - University of Copenhagen
Lecturer

Block 3 2024 Introduction to Representation Theory - University of Copenhagen
Teaching Assistant

Winter 2021 Quantum Information - Perimeter Scholars International (PSI) Program
Teaching Assistant

Winter 2021 Quantum-classical gaps in causal structures - PSI Winter School
Student project co-lead

Fall 2018 Physics 1 - Phys 111 - University of Waterloo
Teaching Assistant